Engineering Standards and Constraints

QUBO-Based Sensor Node Placement for Device-Free Localization

Mohamed Khalil Brik

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Introduction

This document identifies the engineering standards and constraints that govern the design, implementation, and evaluation of the QUBO-based sensor node placement system. It serves as a formal record of the standards followed and the boundaries within which the system was built, as required for the final thesis submission package.

Applicable Engineering Standards

IEEE 830: Software Requirements Specification

The System Requirements and Design Specifications document follows the structure recommended by IEEE Std 830-1998. Requirements are written using consistent identifiers (FR-1 through FR-10, NFR-1 through NFR-5), are individually testable, and are organized by functional area. This ensures the requirements are complete, unambiguous, and traceable to the implementation.

IEEE 829: Software Test Documentation

The Verification and Validation Report follows the test documentation practices of IEEE Std 829-2008. Each test case includes a unique identifier, the requirement being verified, the input conditions, the expected outcome, and the actual outcome. This structure makes the testing process auditable and reproducible.

IEEE 1012: Software Verification and Validation

The overall V&V approach is aligned with IEEE Std 1012-2016. Verification checks that each component was built correctly according to the specification. Validation checks that the full system produces results that are meaningful and consistent with the research goals. Both levels are addressed in the accompanying V&V report.

IEEE 754: Floating-Point Arithmetic

All floating-point computations in the system rely on IEEE 754 double precision (64-bit) arithmetic, which is the default in Python's `numpy` and `pandas` libraries. The binary search tolerance $\epsilon = 0.01$ is chosen to be well above the resolution of double precision arithmetic, avoiding any numerical instability near the convergence boundary.

IEEE Publication Figure Standards

All figures generated for the thesis follow the IEEE journal figure guidelines. Figures use a serif font (Computer Modern), a minimum font size of 9 pt, a line width of at least 0.8 pt, and are saved at 300 DPI. Colors are chosen to remain distinguishable in both color and grayscale print. Figure width is set to 3.5 inches to match a single IEEE journal column.

Reproducible Research Standards

The system follows the reproducibility principles outlined by ACM and IEEE for computational research. All stochastic components use a fixed random seed (seed = 42 for single-run experiments; seeds 0 through 99 for the multi-run experiment). All input data and output results are stored as plain CSV files so that any reviewer can re-run the pipeline and obtain identical results.

Design Constraints

Hardware Constraints

- **No quantum hardware access.** The system was designed to run entirely on classical hardware. The quantum annealing solver is emulated using OpenJij's `SQASampler`, which simulates the behavior of a D-Wave processor. The projected hardware anneal time of 20 μs per read is used for TTS calculations, but no actual D-Wave machine was accessed.
- **No specialized GPU or FPGA.** The system runs on a standard laptop CPU. All matrix operations use NumPy, which may use multi-threaded BLAS internally but does not require a GPU.
- **Sensor node count fixed at $N = 10$.** The environment provides exactly 10 candidate nodes, yielding a finite set of AP-MP pairs. The system is not required to scale beyond this dataset for the thesis.

Software Constraints

- **Python 3.8 or later.** The system uses f-strings, `pathlib`, and type hints that require at least Python 3.8.
- **OpenJij dependency.** The QUBO solvers depend on the OpenJij library. The system does not implement its own annealing algorithm. Any future change to the OpenJij API could require updates to the solver scripts.
- **No internet connection required at runtime.** All data is stored locally. The system does not call any external APIs or download any files during execution.

Budget Constraints

- **AP budget: $n \leq 5$.** At most 5 distinct Access Point nodes may appear in the selected set of links.
- **MP budget: $m \leq 5$.** At most 5 distinct Monitoring Point nodes may appear in the selected set of links.
- These budgets reflect realistic hardware deployment limits and are enforced as hard constraints throughout the binary search. Any solution that violates either budget is immediately rejected.

Algorithmic Constraints

- **Binary search tolerance $\epsilon = 0.01$.** The search terminates when the interval $[a, b]$ shrinks to less than 0.01. This results in at most $\lceil \log_2(1/0.01) \rceil = 7$ iterations.
- **QUBO formulation is unconstrained.** Budget feasibility is not encoded as a penalty term inside the QUBO. This is a deliberate design choice to avoid penalty

weight tuning. The QUBO only balances importance against redundancy through the single parameter α.

- **k-NN with $k = 3$ fixed.** The localization evaluator uses exactly 3 nearest neighbors. This value was chosen to balance between noise sensitivity and spatial resolution and is held constant across all experiments for fair comparison.

Data Constraints

- **Fixed dataset.** The fingerprint RSS data, coordinates, and precomputed importance and redundancy matrices are fixed inputs. The system does not perform any data collection or augmentation.
- **Train/test split is predetermined.** The split between training and test fingerprints is defined in the input CSV files. The system does not shuffle or re-split the data.
- **Importance scores are RSS variance.** The importance of each AP-MP link is defined as the variance of its RSS measurements across fingerprint locations. Links with higher variance carry more discriminative power for localization.
- **Redundancy is measured by RSS correlation.** The redundancy between two links is the Pearson correlation of their RSS vectors. Selecting two highly correlated links adds little new information.

Time Constraints

- **Single-run binary search must complete within 10 minutes** on a standard laptop. In practice, SA with 100 reads and 1000 sweeps per iteration converges in under 2 minutes.
- **The 100-trial multi-run experiment** is intended as an offline batch job and has no strict time limit, though it typically completes within 3 hours on the same hardware.

Ethical and Academic Constraints

- **No human subject data.** The RSS fingerprint dataset does not contain any personally identifiable information. It consists only of signal strength measurements at fixed physical locations.
- **Reproducibility.** All experiments use fixed seeds so that claims made in the thesis can be independently verified from the submitted code and data.
- **Honest reporting.** The greedy baseline is included and honestly compared against the QUBO methods. No results are selectively omitted.

Summary Table

Standard / Constraint	Category	Impact on System
IEEE 830	Documentation	Structure of SRS/SDS document
IEEE 829	Testing	Structure of V&V test cases
IEEE 1012	V&V process	Scope of verification and validation
IEEE 754	Arithmetic	Floating-point stability of binary search
IEEE figures	Visualization	DPI, font size, line width of all figures
Reproducibility	Research	Fixed seeds, plain CSV outputs
AP budget ≤ 5	Hardware	Hard constraint in binary search
MP budget ≤ 5	Hardware	Hard constraint in binary search
$\epsilon = 0.01$	Algorithmic	Binary search convergence bound
$k = 3$ (k-NN)	Algorithmic	Fixed evaluation metric
No real QA hardware	Hardware	SQA emulation via OpenJij
Python 3.8+	Software	Runtime environment
No internet	Software	Fully offline execution
No human data	Ethical	No IRB approval required

Table 1: Summary of applicable standards and constraints.